Analog Guide

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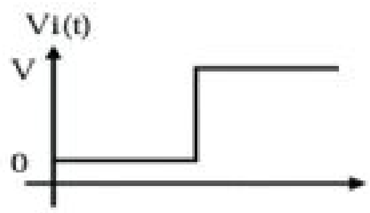
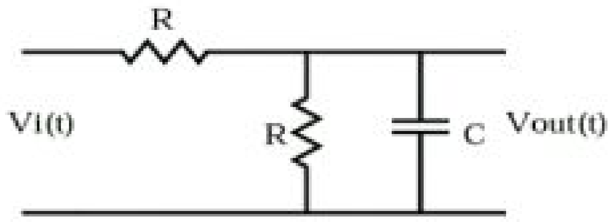
1. Calculate Vout(s)/Vin(s). plot Vo(t). calculate time constant and pole frequency.





$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{\dfrac{R}{1+RCs}}{R + \dfrac{R}{1+RCs}} = \frac{R}{2R + R^2Cs}$$

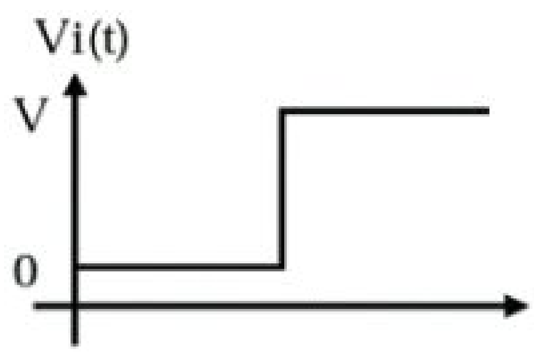
$$= \frac{1}{2 + RCs} = \underline{\underline{\frac{1}{RC\left(s + \frac{2}{RC}\right)}}}$$

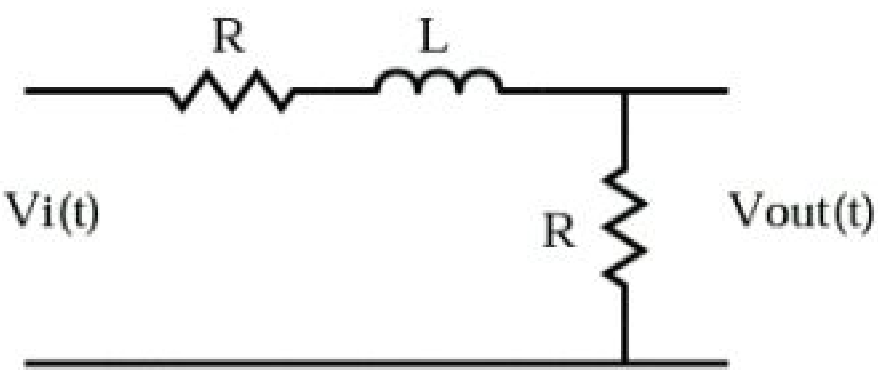
$$V_o(s) = \frac{V_o}{s(2 + RCs)} = \frac{V_o/2}{s\left(1 + \frac{RCs}{2}\right)}$$

$$= \frac{V_o}{2}\left(1 - e^{-t/\tau}\right) \qquad \tau = \frac{RC}{2}$$

pole = $-2/RC$

2. In the following circuit plot iL(t), VL(t)





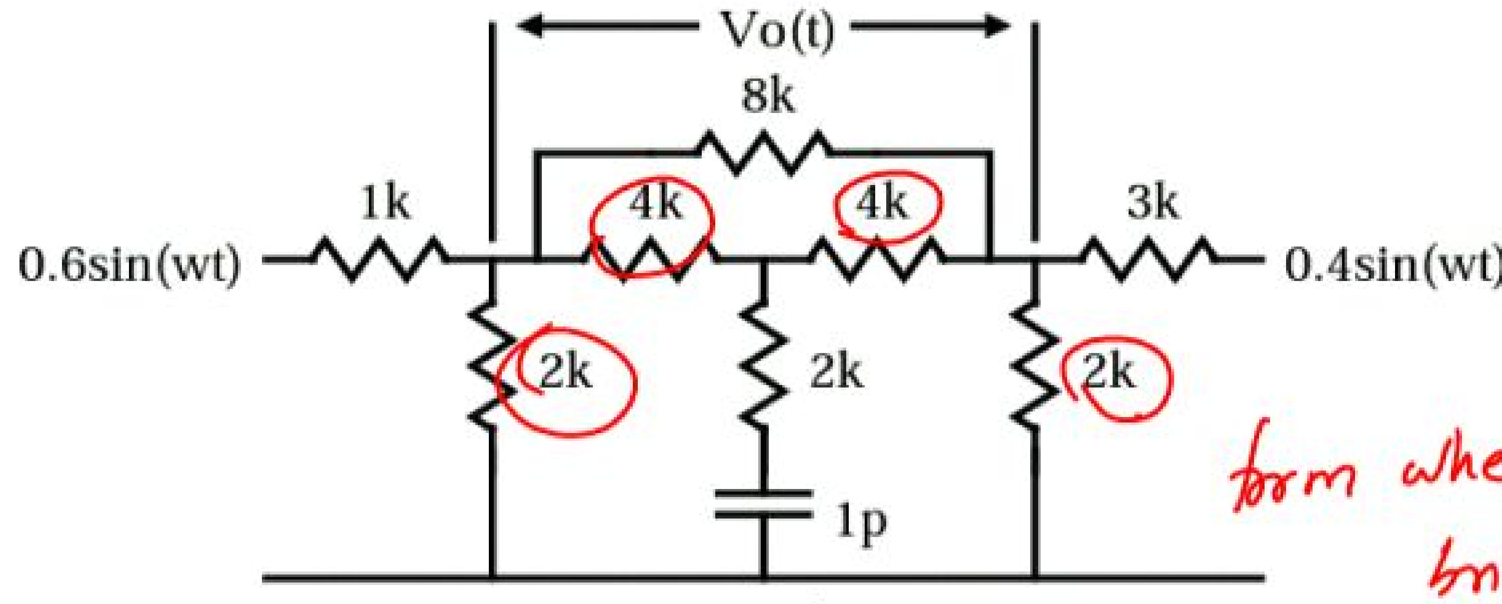
$$V_{out}(t) = \frac{V_0}{2}\left(1 - e^{-t/\tau}\right) \qquad \tau = \frac{L}{2R}$$

$$i_L(t) = \frac{V_0}{2R}\left(1 - e^{-t/\tau}\right)$$

$$V_L(t) = L\frac{di_L(t)}{dt} = L\frac{V_0}{2R}\frac{1}{\tau}e^{-t/\tau}$$

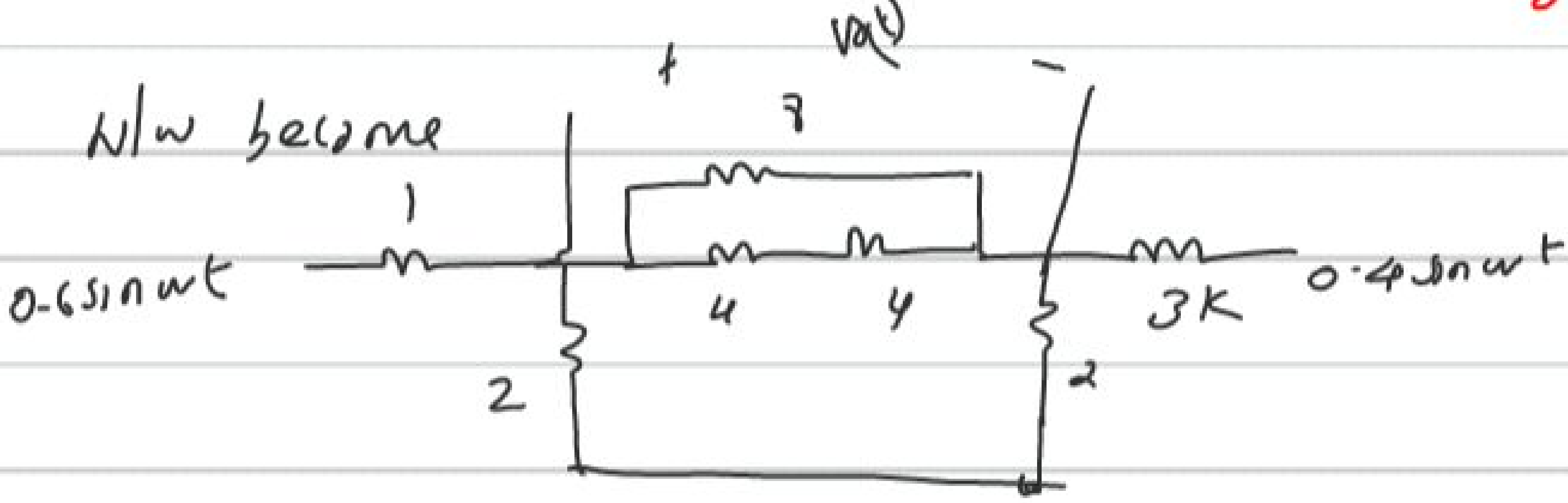
$$= \frac{L V_0}{2R}\frac{2R}{L}e^{-t/\tau} = \underline{\underline{V_0 e^{-t/\tau}}}$$

3. Find Vo(t) in the circuit below.



form wheatstone bridge

N/w become



$= \underline{\underline{1.4}}$

4. A system has an SNR of 60 dB. If an uncorrelated noise of 1 mV is added in a 1V of signal to it, then what is the SNR?

$$SNR_{in} = 10\log\left(\frac{1^2}{(10^{-3})^2}\right)$$

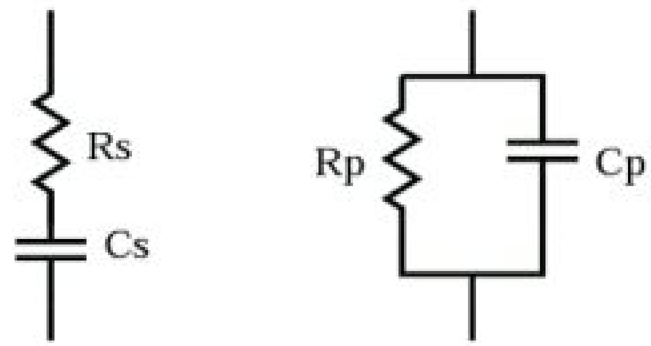
$$= 10\log(10^6) = \underline{\underline{60\,dB}}$$

$$\frac{SNR_{out}}{SNR_{in}} = 10^6$$

$$SNR_{out} = 10^6 \times 10^6 = 10^{12}$$

$$\Rightarrow \underline{\underline{120\,dB}}$$

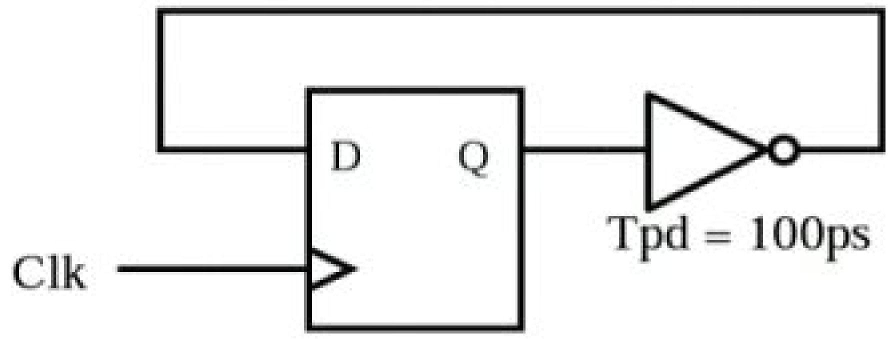
5. Both ckts are equivalent. Express Rp and Cp interms of Rs and Cs. Find ω range for which these both are equivalent. Assume high quality factor.



$$R_p = R_s (1+Q^2)$$

$$C_p = \frac{C_s Q^2}{1+Q^2}$$

6. Calculate the maximum clock frequency.



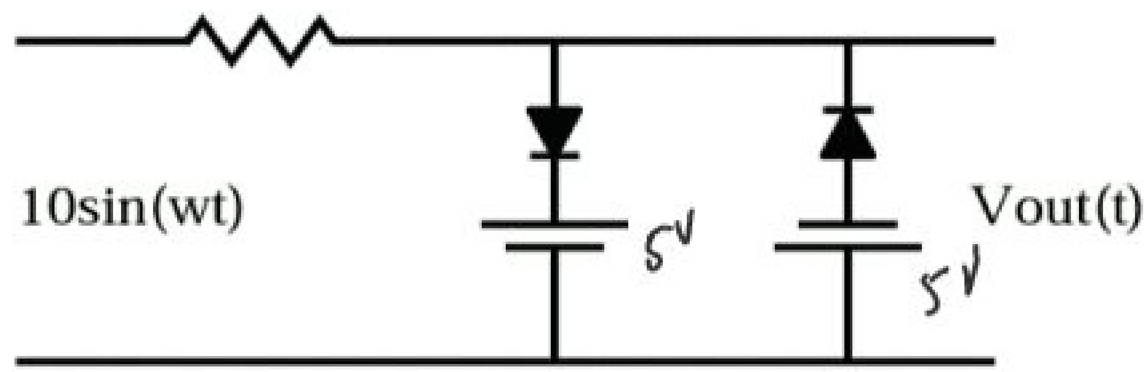
Tcq = 1ns
Tsetup = 200ps
Thold = 300ps

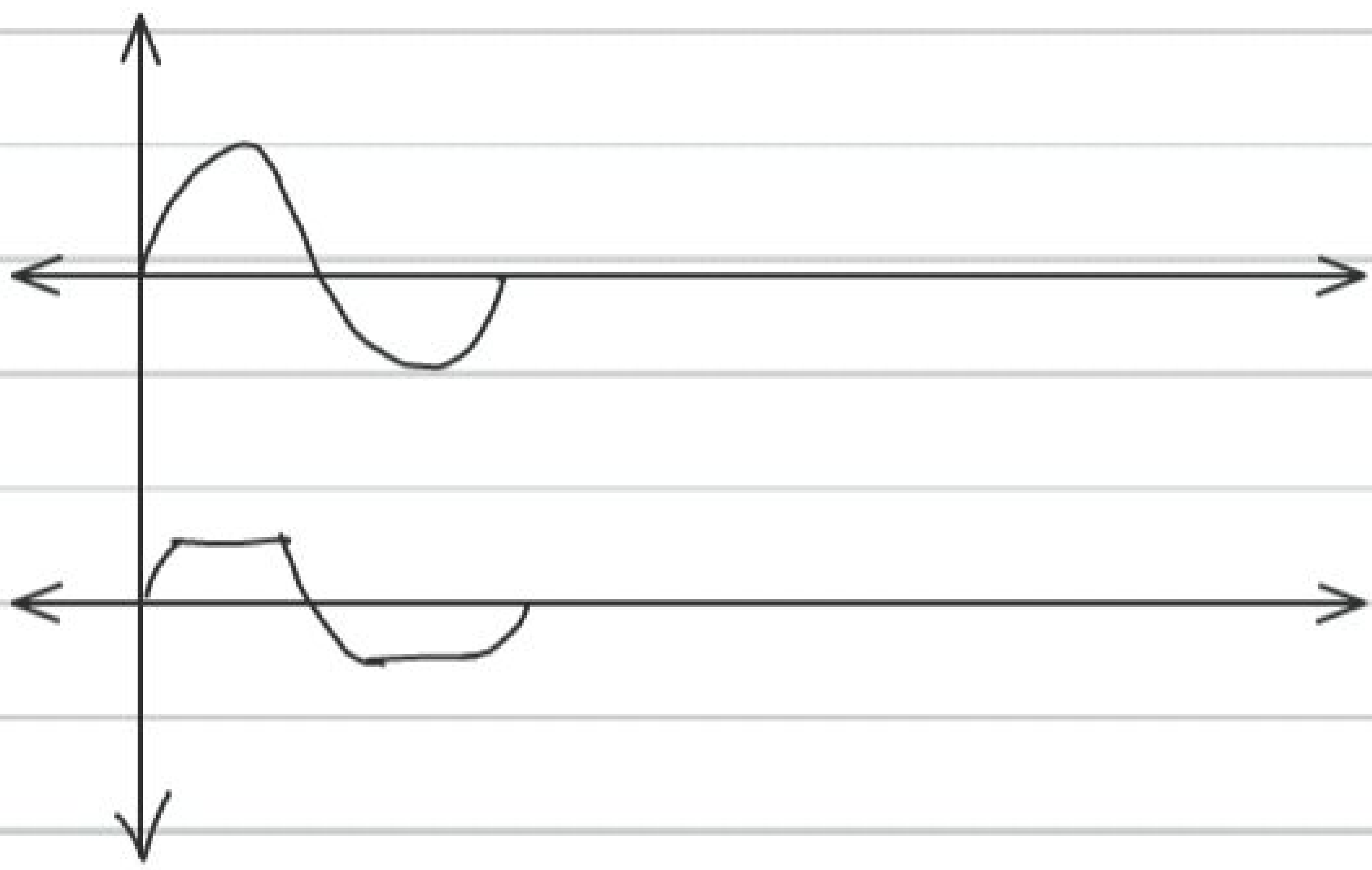
$T \geqslant t_{cq} + t_{logic} + t_{setup}$

$\geqslant 1 + 0.1 + 0.2 = \underline{\underline{1.3}}\ n$

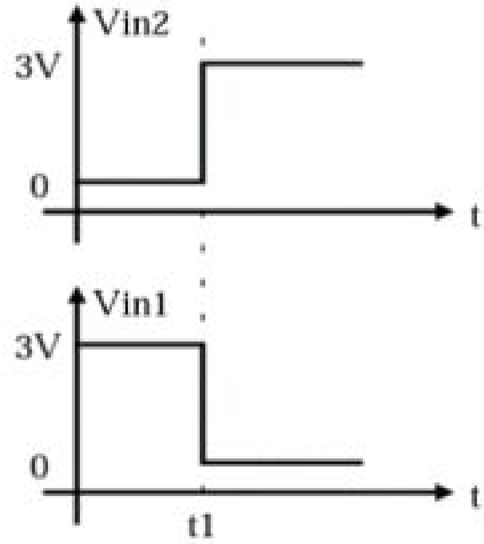
7. Plot Vout.

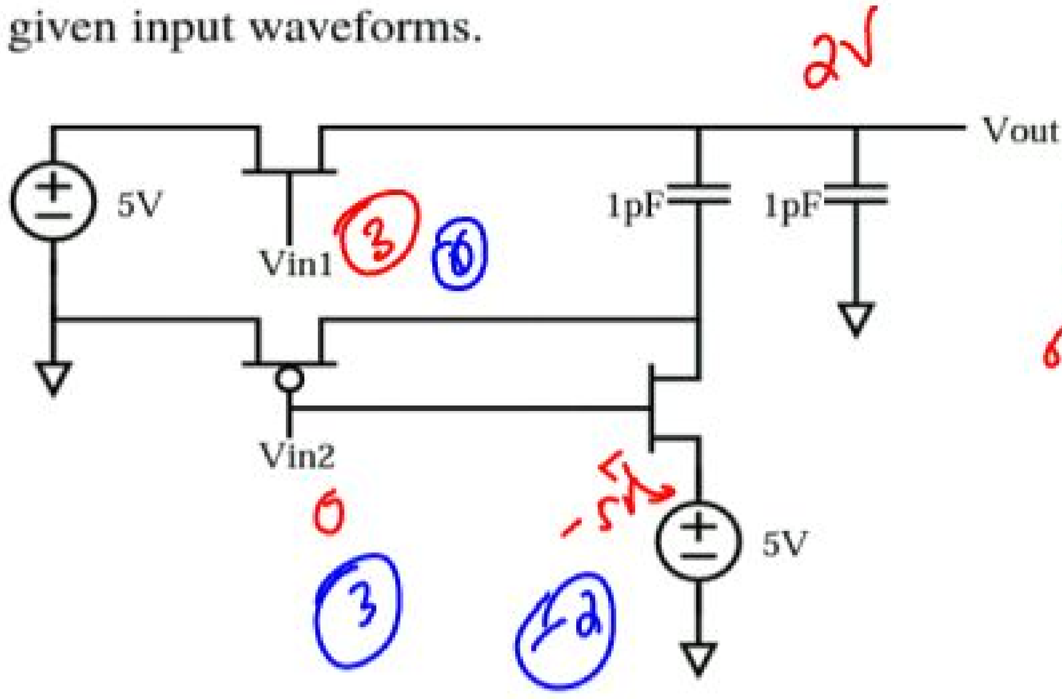
Diod drop ≈ 0

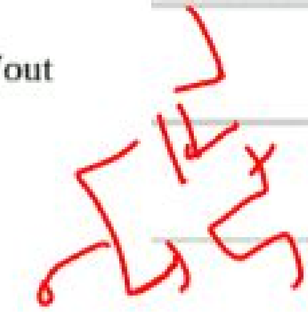




8. Plot Vout with respect to the given input waveforms.

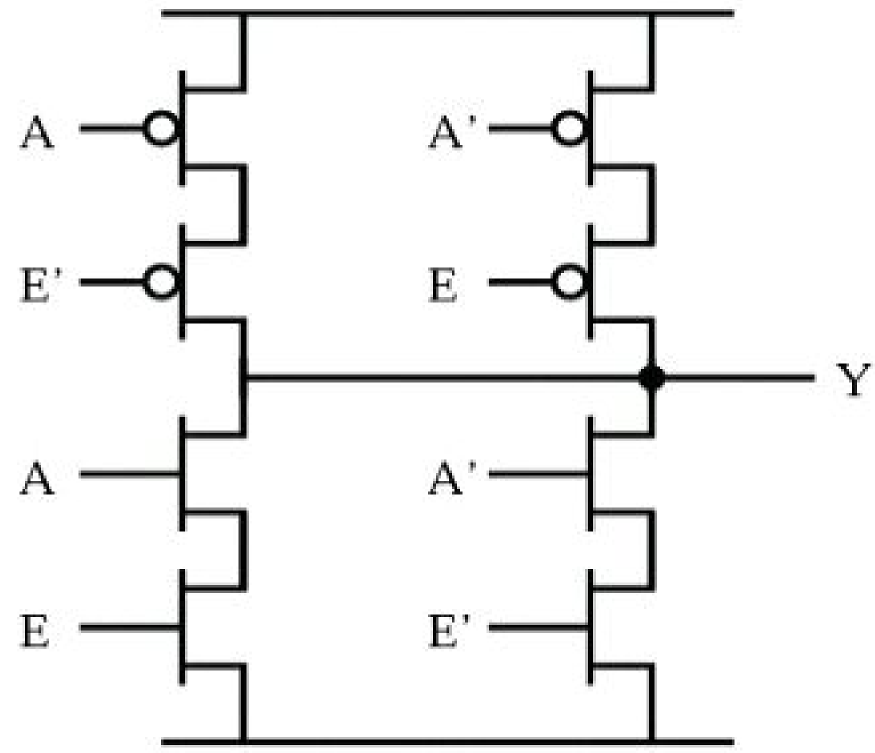






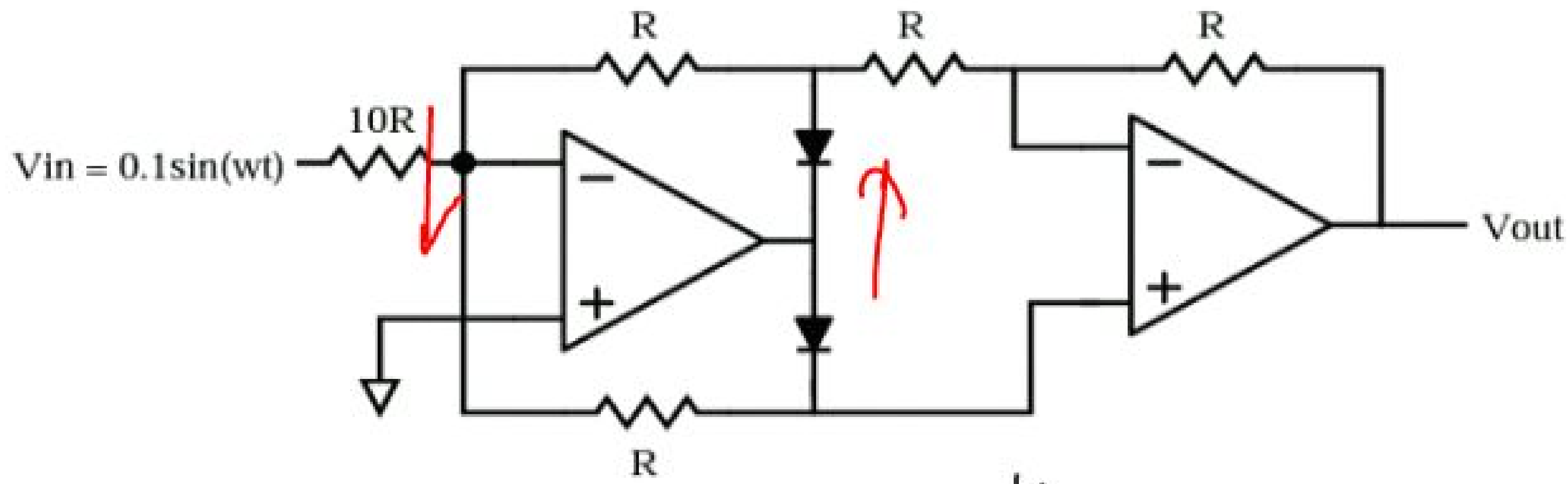
9. Design a divide-by-3 counter using D-flipflops. The duty cycle of the divided clock should be 2/3.

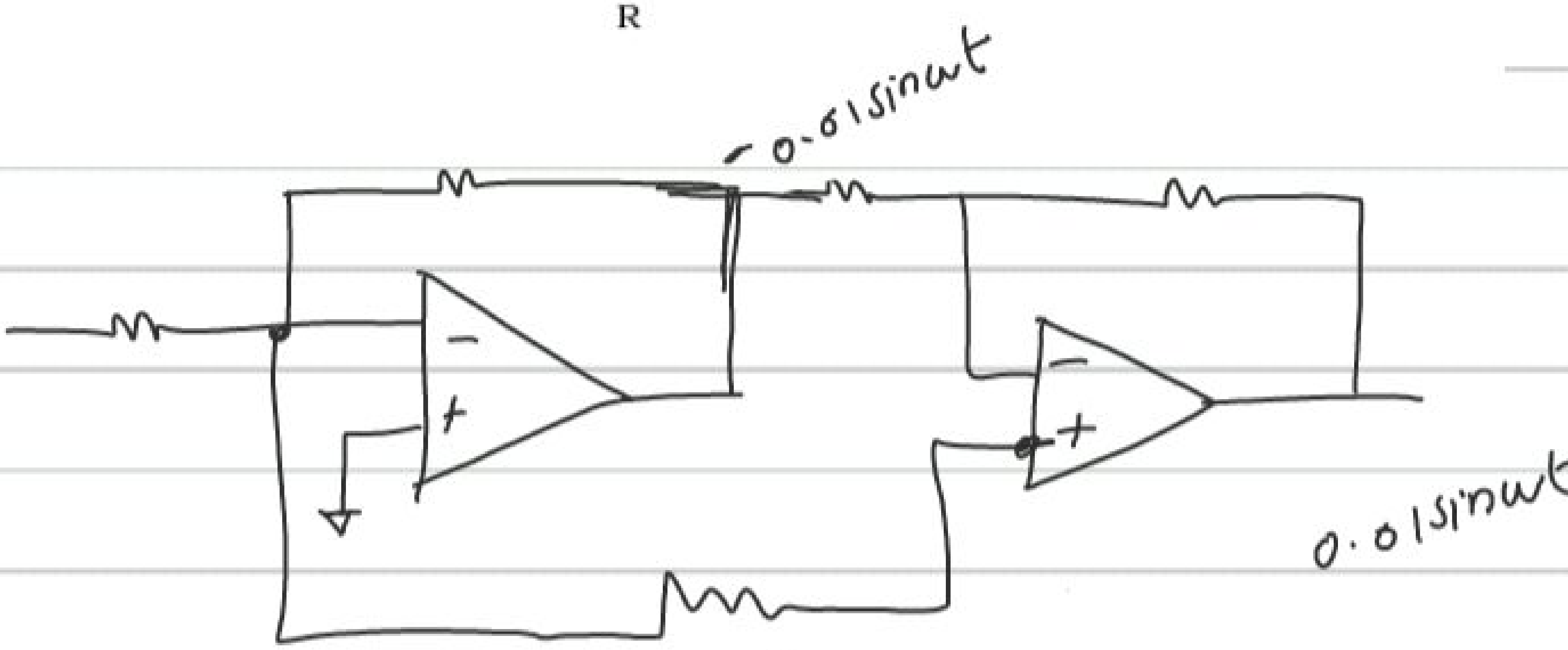
10. What is the function of the following circuit?

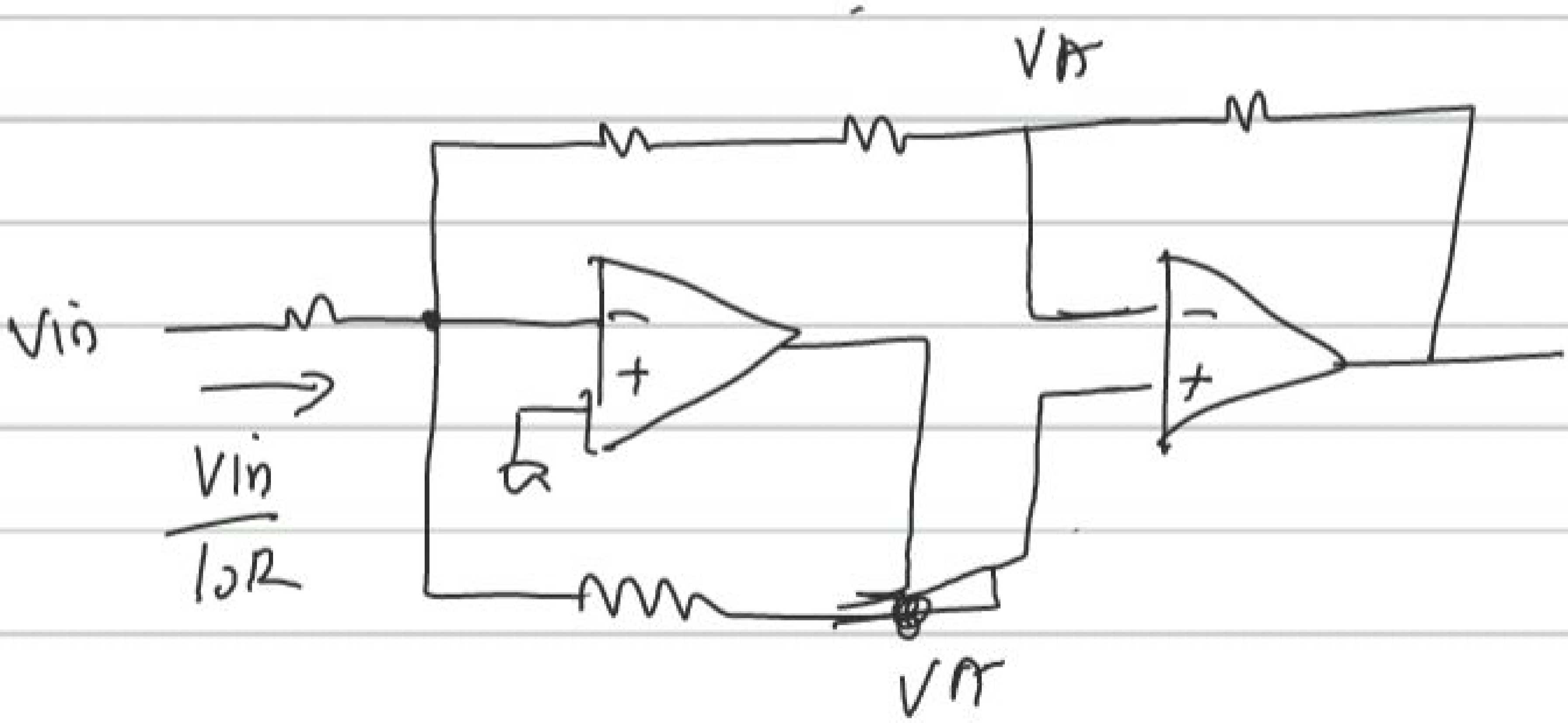


XOR gate

11. Plot the output Vout.







$$-\frac{VA}{2R} + -\frac{VA}{R} = \frac{Vin}{10R}$$

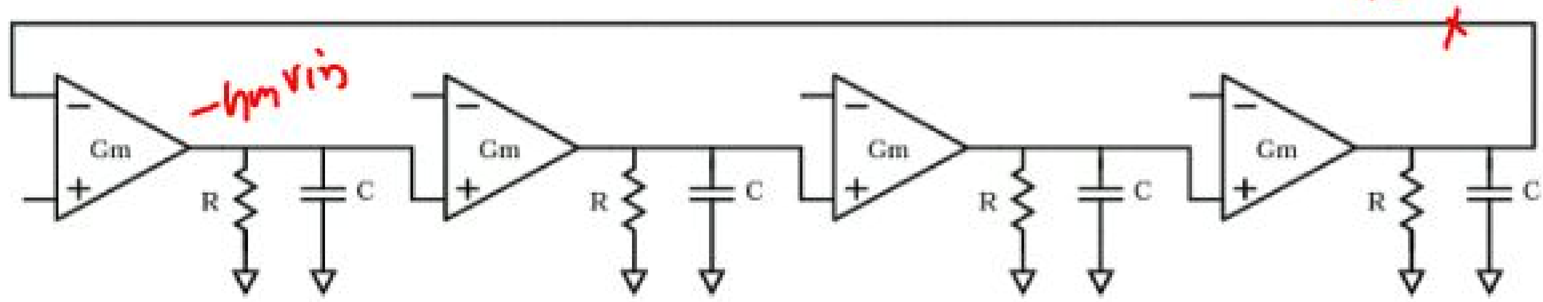
$$1.5\,VA = -\frac{Vin}{10}$$

$$VA = \frac{-Vin}{15}$$

$$\frac{-V_{in}}{2R \parallel 5} = \frac{V_{out} - V_{in}}{R}$$

$$\frac{V_{in}}{30R} + \frac{V_{in}}{R} = \frac{V_{out}}{R}$$

12. Calculate the frequency of oscillation. What is the minimum required Gm for oscillation?



$$-\left(\frac{g_m R}{1+RCs}\right)^4 = 1$$

$$-(g_m R)^4 = (1+RCs)^4$$

$$= \left(1 + R^2C^2s^2 + 2RCs\right)^2$$

$$= 1 + R^4C^4s^4 + 4R^2C^2s^2 +$$

$$2R^2C^2s^2 + 4R^3C^3s^3 + 4RCs$$

$$1 + (g_m R)^4 + R^4C^4\omega^4 - 6R^2C^2\omega^2 - j\omega^3R^3C^3 4$$

$$j4\omega RC = 0$$

$$\omega^3 R^3 C^3 = \omega RC$$

$$\omega^2 = \frac{1}{R^2C^2}$$

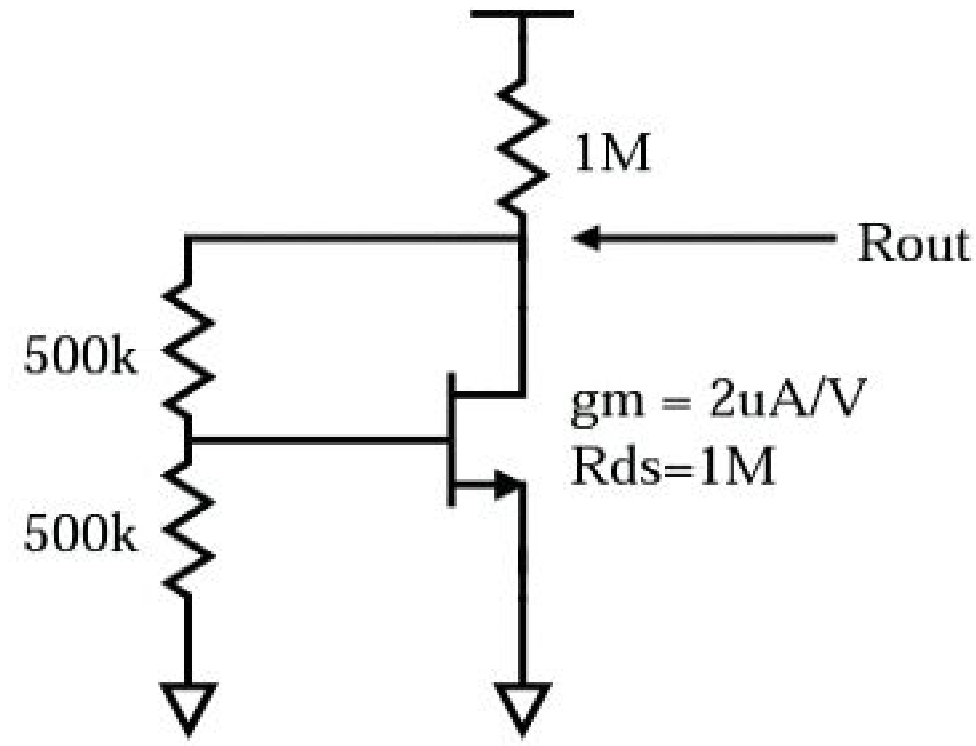
$$\omega = \frac{1}{RC}$$

$$\left|\frac{\omega mR}{1+j1}\right|^4 = 1$$

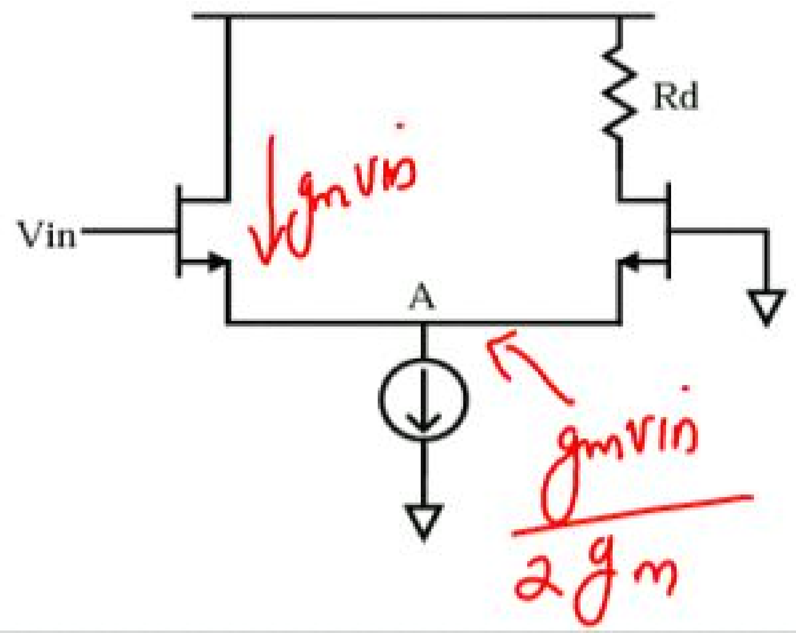
$$\left|\frac{\omega mR}{\sqrt{2}}\right|^4 = 1$$

$$\underline{\underline{\omega mR = \sqrt{2}}}$$

13. Calculate the output impedance.

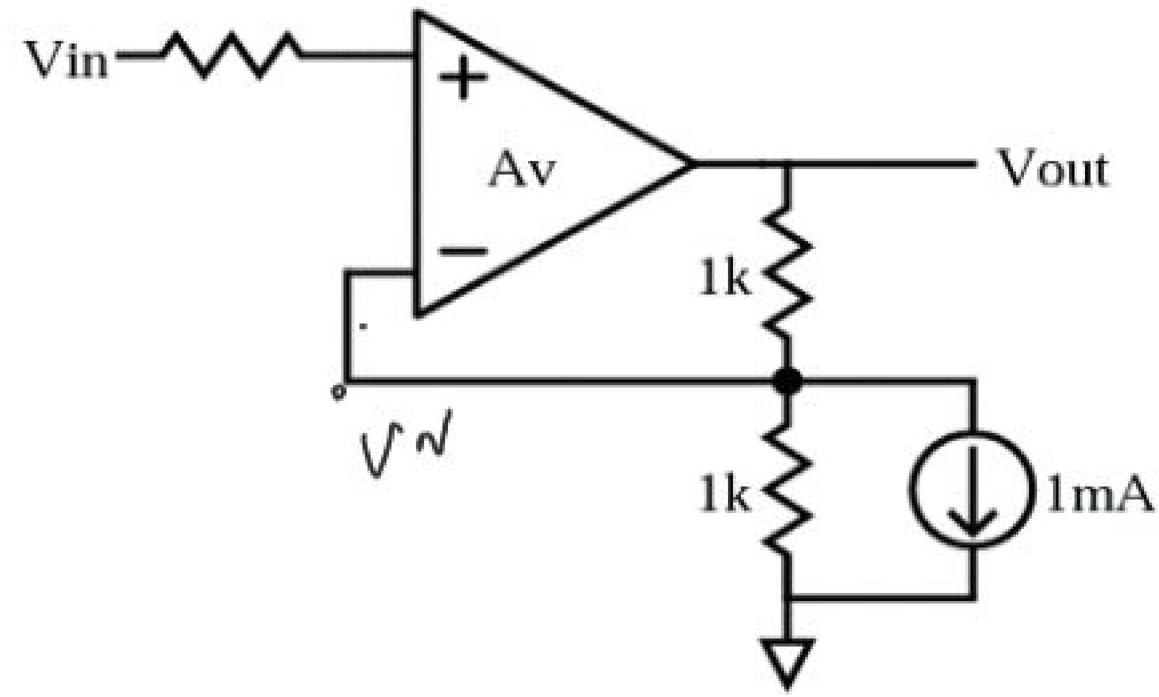


14. Both transistors are biased in saturation. Calculate “VA/Vin”. Neglect the body effect.



$$\frac{VA}{Vin} = 1/2$$

15. Calculate the output if (i) gain Av = infinity (ii) gain Av = 10



(i)

$$\frac{Vin}{1K} + 1mA + \frac{Vin - Vout}{1k} = 0$$

$$2vin + 1 = Vout$$

(ii)

$$VN + 1 + VN - Vout = 0$$

$$Vout = 2VN + 1$$

$$A(Vin - VN) = Vout$$

$$Vin - VN = \frac{Vout}{10}$$

$$VN = Vin - \frac{Vout}{10}$$

$$Vout = 2\left(Vin - \frac{Vout}{10}\right) + 1$$

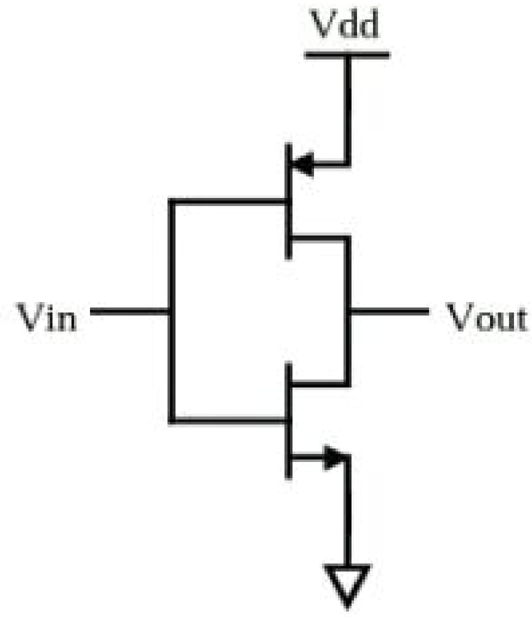
$$= 2Vin + 1 - \frac{Vout}{5}$$

$$\left(1+\frac{1}{5}\right) v_{out} = 2v_{in} + 1$$

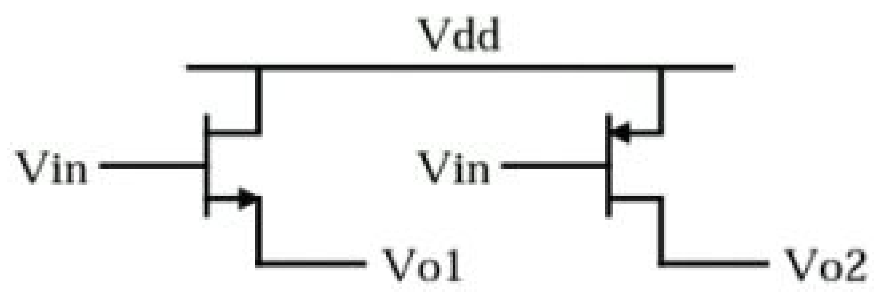
$$6/5\ v_{out} = 2v_{in} + 1$$

$$v_{out} = \frac{10}{6} v_{in} + 5/6$$

16. If the following inverter biased in the middle of Vdd, what is the small signal gain? (Answer: $g_m r_o$)

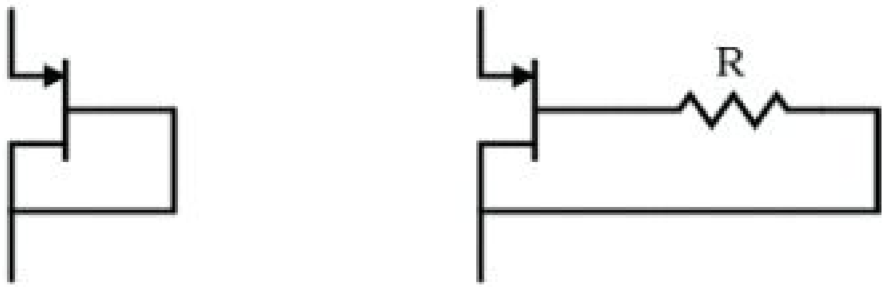


22. For the following circuits, Vdd=5v, tell me what are Vo1 and Vo2 when Vin is 5V, 3V, 2.5V and 0V.



Vin = 5V

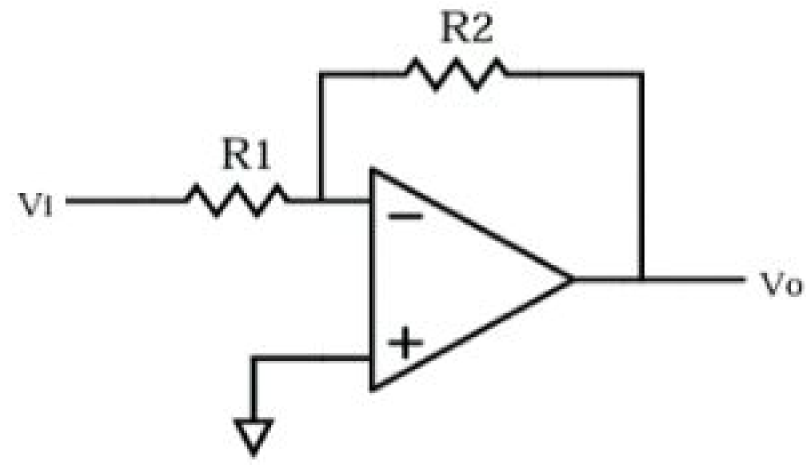
23. What are the effective resistance from source to drain of the following two transistors? (The value of the resistance is R). Answer: both of them are 1/gm.



(a) (b)

Both $1/gm$

24. In the following figure, if the two resisters are equal, what is its -3dB bandwidth? Compare its stability with that of a source follower.



$$w_{3dB} = w_u \times \frac{R_1}{R_1 + R_2}$$

w_u = unity gain B.W

25. For the following circuit, if the input is a rail-to-rail square wave, plot the wave after the inverter and vo.